

Custom Lightweight Discovery Protocol (CLDP) Specification

1 Introduction

The Custom Lightweight Discovery Protocol (CLDP) is a simple network protocol implemented using raw sockets in POSIX C. It allows nodes in a closed network to announce their presence and query each other for metadata, such as system time, hostname, and CPU load.

2 CLDP Packet Format

Each CLDP packet consists of a fixed-size header followed by an optional payload.

2.1 Header Format (8 bytes)

Field	Size (Bytes)	Description
Message Type	1	Defines the type of message (HELLO, QUERY, RESPONSE)
Payload Length	1	Length of the payload in bytes
Transaction ID	2	Unique ID for matching requests and responses
Reserved	4	Unused (set to 0)

2.2 Message Types

Type	Value	Description
HELLO	0x01	Announces the presence of a node
QUERY	0x02	Requests metadata from nodes
RESPONSE	0x03	Responds with requested metadata

3 Supported Metadata

The CLDP protocol supports the following metadata queries:

1. **Hostname** - The network name of the responding node.
2. **System Time** - The current system time of the responding node.
3. **CPU Load** - The CPU utilization of the responding node.

4 Communication Flow

1. Servers periodically send **HELLO** messages every 10 seconds to announce their presence.
2. Clients send **QUERY** messages to discover available servers and request metadata.
3. Servers respond with **RESPONSE** messages containing requested metadata.

5 Implementation Details

- Uses raw sockets (`AF_INET`, `SOCK_RAW`).
- IP header is manually crafted (Protocol Number: 253).
- Messages are parsed at the IP level to extract CLDP packets.
- Clients and servers communicate without transport-layer protocols (TCP/UDP).